

PRD: AI-Powered Credit Card Recommendation Engine

Product: FreechargeBiz × Axis Bank Credit Card Recommendation Engine

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Table of Contents

1. [Problem Statement](#)
 2. [User Personas](#)
 3. [Core Features & User Journey](#)
 4. [Technical Architecture & Data Strategy](#)
 5. [User Stories & Prioritization](#)
 6. [Future Roadmap](#)
 7. [Appendix](#)
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1. Problem Statement

1.1 Background

India has **118.63 million active credit cards** (March 2026, RBI) with FY26 total spend hitting **₹23.62 trillion**, a +12% YoY jump. Usage is shifting from offline/small-ticket to **digital-first, high-value, lifestyle-driven** spend, with e-commerce now accounting for 64% of monthly card volume. Banks are moving from aggressive issuance to quality customer acquisition.

Despite this scale, the card **discovery and selection experience is broken**.

1.2 The Problem

When a user decides they want a credit card, they face:

- **Information overload:** 50+ card variants across issuers with overlapping features and jargon-heavy benefit structures

- **Generic recommendations:** most fintech platforms surface cards based on issuer commission, not user fit
- **High drop-off on forms:** average fintech onboarding drop-off is **63–68%**; 70% of financial institutions lost clients in 2025 due to inefficient onboarding (Fenergo, 2025)
- **No behavioral signal:** recommendations ignore actual spending patterns, income proxies, or lifestyle context
- **Post-selection regret:** users frequently discover their card doesn't match real spending (e.g., a travel card for someone who rarely flies)

1.3 The Business Problem

FreechargeBiz sits at a strategic intersection: access to user transaction signals (via UPI/SMS data) and a distribution partnership with Axis Bank. Despite this, card recommendations are not contextually driven, resulting in:

- Low conversion rates on card application pages
- High abandonment after recommendation due to poor relevance
- Missed revenue from eligible users who exit due to poor UX or irrelevant suggestions

1.4 The Opportunity

An AI-powered recommendation engine that maps a user's **financial footprint** (SMS data, bureau signals, spending categories, income proxies) to the right Axis Bank / Freecharge card can:

- Deliver the **right card to the right user** in under 60 seconds
- Surface personalized rationale: *"This card gives you 5% back on Swiggy; you spent ₹4,200 there last month"*
- Increase application conversion through trust and relevance
- Reduce time-to-recommendation and form drop-off significantly

1.5 Core Flow

```
graph TD; A[User opens FreechargeBiz app / lands on card discovery] --> B[Consent screen: user grants access to SMS / financial data]; B --> C[AI engine parses spending signals + income proxy]; C --> D[Bureau check (soft pull): eligibility scoring]; D --> E[Recommendation engine maps user profile → best-fit card(s)]; E --> F[Personalized recommendation page: card shown with user-specific rationale]; F --> G[User applies → redirected to Axis Bank application flow]; G --> H[Application submitted → confirmation + next steps];
```

User opens FreechargeBiz app / lands on card discovery

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Consent screen: user grants access to SMS / financial data

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AI engine parses spending signals + income proxy

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Bureau check (soft pull): eligibility scoring

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Recommendation engine maps user profile → best-fit card(s)

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Personalized recommendation page: card shown with user-specific rationale

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User applies → redirected to Axis Bank application flow

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Application submitted → confirmation + next steps

1.6 Scope

In scope (v1):

- Users on the FreechargeBiz app who express intent to explore credit cards
- Axis Bank credit card catalog (full portfolio, not just Freecharge co-branded variants)
- SMS-based spending signal parsing + explicit user inputs as data inputs
- Single recommendation surface: one primary card + up to 2 alternatives

Out of scope (v1):

- Full bureau report ingestion (soft pull eligibility only in v1)
- Cross-issuer recommendations (non-Axis Bank cards)
- Existing credit card holders looking to upgrade (future iteration)
- Offline / branch-based application flows

Card catalog note: The Freecharge co-branded portfolio has 2 variants: the **Axis Bank Freecharge Credit Card** (₹250 joining fee, EDGE Reward Points) and the **Axis Bank Freecharge Plus Credit Card** (invite-only, 5% cashback on Freecharge spends, 2% on Ola/Uber). The engine must draw from the **broader Axis Bank portfolio** to serve diverse user profiles.

2. User Personas

Attribute	Riya, Digital Millennial	Ramesh, Salaried Professional	Ananya, First-Timer	Vivek, Business Traveler
Age	27	38	22	44
Occupation	UX Designer, Bangalore startup	Branch Manager, PSU Bank, Pune	Junior SWE, 6 months exp	Regional Sales Head, FMCG
Monthly Income	₹85,000	₹90,000	₹45,000	₹2,50,000 (salary + variable)
Existing Cards	None	1 SBI co-branded (rarely used)	None	HDFC Infinia, ICICI Sapphire
Top Spend Categories	Food delivery, OTT, weekend travel	Fuel, groceries, utilities, EMIs	Online shopping, gaming, food delivery	Air travel, hotels, business dining
Pain Point	Overwhelmed by options; doesn't know where to start	Distrusts "too good to be true" offers; wants clarity on fees	No credit history; feels excluded from premium options	Wants to know if Axis card genuinely beats his current stack

Goal	Lifestyle-matched rewards, no hidden fees	Clear cashback on everyday spend	First card to start building credit score	Quantified value vs. existing cards
Tech Comfort	High	Medium	Very High	High
Key Behavioral Insight	Abandons form-like flows; responds to social proof and personalized framing	Won't apply without understanding annual fee payoff; needs trust signals	Needs positive framing: <i>"Start here, upgrade in 12 months"</i>	Only converts on hard numbers: <i>"₹18,000 more in rewards annually"</i>
Engine Behavior	Standard recommendation, lifestyle card focus	Cashback / fuel card routing	Thin bureau path → entry-level card options	Premium card matching + value comparison vs. existing cards

3. Core Features & User Journey

3.1 Core Features

Feature	Description	Priority
Consent & Data Collection	Transparent SMS permission screen with a fallback explicit input form (4 questions max)	Must Have
SMS Signal Parser	Parses financial SMS to extract income proxy, spend categories, and EMI obligations	Must Have
Soft Bureau Check	Pulls credit eligibility signal without impacting user's CIBIL score	Must Have
Recommendation Engine	Maps user profile to best-fit Axis Bank card; ranks by benefit-to-spend overlap	Must Have

Personalized Recommendation Page	Shows primary card with user-specific rationale + up to 2 alternate cards	Must Have
Apply Now Redirect	Deep-links to Axis Bank application with pre-filled data where API permits	Must Have
Alternate Card Suggestions	Shows up to 2 alternative cards below the primary recommendation	Should Have
Comparison Mode	Side-by-side comparison of 2–3 shortlisted cards	Could Have
Save & Return	Bookmarks recommendation for users who want to apply later	Could Have

3.2 User Journey

User taps "Find my perfect card" (home feed / post-recharge / bill payment screen)

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Consent Screen: explains what data is read and what is not stored

→ Path A: SMS access granted → parser runs in background (~10 sec)

→ Path B: Skipped → 4-question explicit input form

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Soft bureau check runs in parallel

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Spending profile built + eligibility scored

→ If thin bureau (score < 650) → routed to entry-level card options

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Recommendation engine selects best-fit card

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Recommendation Page

- Primary card with "Why this card for you" (3 spend-linked callouts)
- Annual fee vs. estimated reward value
- Up to 2 alternate cards

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User taps "Apply Now" → Axis Bank application flow (pre-filled where permitted)

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Application submitted → confirmation screen

3.3 Drop-off Risk

Stage	Risk	Mitigation
Consent Screen	High	Clear copy on what's read vs. stored; fallback path always visible
Input Form (Path B)	Medium	Max 4 questions, range sliders, conversational tone
Apply → Axis redirect	Medium	Pre-fill data, set expectation before leaving the app

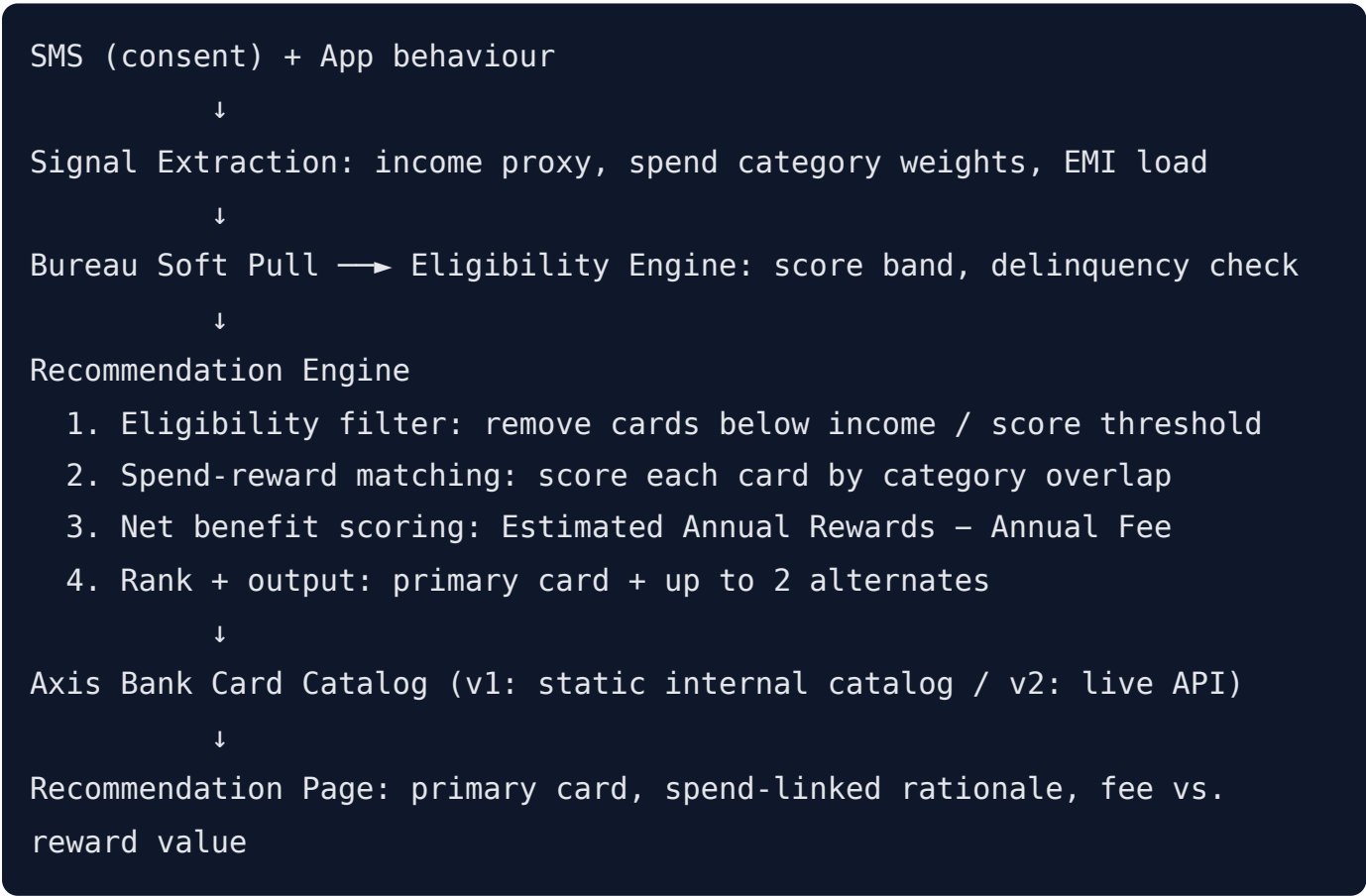
4. Technical Architecture & Data Strategy

4.1 Data Inputs

Source	Signals Extracted	Path
SMS Parser	Income proxy (salary credits), spend categories (merchant SMS), EMI load, utility payments	Primary (consent granted)
Bureau API: Soft Pull	Credit score band, active credit products, delinquency flags	All paths
Explicit User Inputs	Income range, top spend categories, card goal, existing card status (4 questions max)	Fallback (SMS denied)
FreechargeBiz Platform	Bill payment categories, recharge patterns, UPI spend frequency	Supplementary (all paths)

SMS text is processed on-device where possible; only structured signals (spend categories, income proxy) are passed to the backend. Raw SMS content is never stored.

4.2 Data Flow



4.3 Recommendation Logic

Step	Logic
Eligibility Filter	Hard removes: min income, min credit score, no delinquency in 12 months
Spend Matching	Maps user's top spend categories to card reward categories; higher overlap = higher score
Net Benefit Score	Estimated Annual Rewards – Annual Fee : powers the "You'd earn ₹X" callout
Final Ranking	Top score = primary card; 2nd & 3rd = alternates (must differ in category); thin bureau → entry-level override

4.4 Data Privacy & Compliance

- SMS parsing on-device; raw text never stored or transmitted
- Soft pull disclosed to user: explicitly stated it does not affect CIBIL score
- Only structured signals retained; user can request full data deletion
- Aligned with RBI Digital Lending Guidelines and Account Aggregator (AA) framework

4.5 Model Architecture & Optimization

Layer 1: SMS Feature Extraction

The SMS parser converts raw transaction messages into a structured user spend profile:

Input	Extraction Method	Output
Salary credit SMS	Regex: detects credit keywords + amount	Monthly income proxy
Merchant transaction SMS	Merchant dictionary lookup (e.g., Swiggy → Food, HPCL → Fuel)	Spend category tag + amount
EMI deduction SMS	Regex: detects EMI / loan keywords	Monthly EMI obligation
Utility / bill SMS	Category dictionary (electricity, broadband, insurance)	Bill spend signal

Sample output of Layer 1:

Income Proxy: ₹85,000/month
Spend Weights: Food 38% | Travel 22% | Fuel 15% | Shopping 18% | Bills 7%
EMI Load: ₹8,000/month

v1 uses regex + merchant dictionary (no training data needed). v2 upgrades to a lightweight NLP text classifier (TF-IDF or DistilBERT) to handle unknown merchants and improve category accuracy.

Layer 2: Content-Based Card Scoring

Each eligible card is scored against the user's spend profile using the following three steps:

Step 1: Monthly reward per category

$$\text{Category Reward} = \text{Spend Weight} \times \text{Monthly Income} \times \text{Card Reward Rate}$$

Step 2: Annual reward estimate

$$\text{Estimated Annual Rewards} = \sum (\text{Spend_Weight}_i \times \text{Income} \times \text{Reward_Rate}_i) \times 12$$

Step 3: Net Benefit Score (cost function)

$$\text{Net Benefit Score} = \text{Estimated Annual Rewards} - \text{Annual Fee}$$

Example: same user, two cards

	Card A: Axis Flipkart	Card B: Axis Vistara
Food (38%)	$38\% \times ₹85K \times 5\% = ₹1,615/\text{mo}$	$38\% \times ₹85K \times 2\% = ₹646/\text{mo}$
Travel (22%)	$22\% \times ₹85K \times 2\% = ₹374/\text{mo}$	$22\% \times ₹85K \times 6\% = ₹1,122/\text{mo}$
Fuel (15%)	$15\% \times ₹85K \times 1\% = ₹127/\text{mo}$	$15\% \times ₹85K \times 1\% = ₹127/\text{mo}$
Shopping (18%)	$18\% \times ₹85K \times 5\% = ₹765/\text{mo}$	$18\% \times ₹85K \times 1\% = ₹153/\text{mo}$

Est. Annual Rewards	₹34,578	₹24,582
Annual Fee	₹500	₹3,000
Net Benefit Score	₹34,078 ✓	₹21,582

Card A wins because the user's heaviest spend categories (food, shopping) align with Card A's reward structure. Card B would win for a user whose primary spend is travel.

Step 4: Final ranking

- Primary card = highest Net Benefit Score among eligible cards
- Alternate cards = 2nd and 3rd ranked (must differ in primary reward category)
- Thin bureau override = entry-level cards surfaced regardless of score

v1 → v2 Upgrade Path

	v1 (Current)	v2 (Future)
SMS Parsing	Regex + merchant dictionary	NLP text classifier (DistilBERT)
Recommendation	Content-based Net Benefit scoring	XGBoost trained on `P(user applies
Cost Function	Maximize Net Annual Benefit	Minimize Binary Cross-Entropy on application conversion
Training Data	Not required	User profile × card pairs + application outcome (apply: yes/no)
Evaluation Metric	Net benefit accuracy	AUC-ROC, application conversion rate, approval rate

5. User Stories & Prioritization

5.1 User Stories & Acceptance Criteria

US-01: As a user, I want to grant SMS access so that the engine analyses my spending without a long form.

- SMS prompt explains what is and isn't stored
- Both grant and deny paths lead to a recommendation
- Parser completes within 10 seconds

US-02: As a user who denied SMS access, I want to answer a few questions so that I still get a relevant recommendation.

- Maximum 4 questions; completable in under 90 seconds
- Income via range slider; spend categories via multi-select chips
- Progress indicator visible throughout

US-03: As a user, I want to see a recommendation tied to my spending so that I can trust the suggestion.

- Primary card displayed with minimum 3 spend-linked callouts
- Each callout references a detected or user-selected spend category

US-04: As a user, I want to know how much I'd earn annually so that I can make an informed decision.

- Net annual reward estimate shown on recommendation page
- Calculated as $\Sigma(\text{spend weight} \times \text{income} \times \text{reward rate}) \times 12 - \text{annual fee}$
- Gross reward and annual fee shown separately

US-05: As a user, I want to apply without a disjointed experience so that the process feels seamless.

- Apply Now CTA deep-links to Axis Bank application flow
- Transition screen shown before leaving the app
- Pre-filled fields passed where Axis Bank API permits

US-06: As a first-timer with no credit history, I want cards I can actually get so that I don't feel rejected.

- Bureau detects thin profile (score < 650 or no history)
- Engine routes to entry-level cards only
- Recommendation copy is forward-looking, never rejection-framed

US-07: As a user, I want to see alternate card options so that I can choose if the primary doesn't feel right.

- Up to 2 alternates shown below the primary card

- Each displays name, top benefit, and annual fee
- Alternates differ in primary reward category from each other and the primary card

5.2 MoSCoW Prioritization Matrix

Priority	Stories	Rationale
Must Have	US-01, US-02, US-03, US-04, US-05, US-06	Core flow: product cannot function without these
Should Have	US-07	Boosts conversion confidence; low additional build effort
Could Have	Save & Return, Comparison Mode	Useful for power users; post-MVP backlog
Won't Have (v1)	Cross-issuer cards, full bureau report ingestion	Out of scope: see Section 1.6

5.3 Component Status

Component	Status	Notes
FreechargeBiz SMS access permission	Existing	Platform already requests SMS permission for other features
UPI / bill payment signal aggregation	Existing	Available from app transaction history
Bureau API (soft pull)	New	Integration with CIBIL / Experian required
SMS signal parser	New	Regex + merchant dictionary: v1 build
Spend profile builder	New	Feature extraction layer
Recommendation engine (scoring)	New	Content-based Net Benefit scorer
Axis Bank card catalog	New: static v1	Live API integration is v2
Recommendation page UI	New	Card display + spend-linked rationale

Apply Now deep-link	New	Axis Bank application redirect
Thin bureau routing logic	New	Entry-level card override path

6. Future Roadmap

Feature	Description	Phase
NLP SMS Classifier	Replace regex parser with DistilBERT/TF-IDF to handle unknown merchants	v2
XGBoost Conversion Model	Train ML model on application data to predict P(apply)	v2
Live Axis Bank Card Catalog API	Replace static catalog with real-time API for live fees and offers	v2
Save & Return	Bookmark recommendation; re-check eligibility on return after 30 days	v2
Card Comparison Mode	Side-by-side comparison of up to 3 cards on reward rate, fee, perks	v2
Spending Personality Archetypes	Cluster users into archetypes (Foodie, Traveler, Saver) for richer cold-start	v2
Cross-Issuer Recommendations	Expand catalog beyond Axis Bank to other issuers	v3
Account Aggregator (AA) Integration	Replace SMS parsing with AA framework for richer, consent-based financial data	v3
Progressive Profiling	Build user financial profile across sessions; each app interaction adds a signal	v3
Real-Time Offer Injection	Surface active limited-time card promotions and fee waivers in recommendation	v3

Appendix

Appendix A: Market Data Reference

Metric	Figure	Period
Total active credit cards	118.63 million	March 2026
YoY card base growth	+7.96%	FY26
Total credit card spend	₹23.62 trillion	FY26
YoY spend growth	+12%	FY26 vs FY25
Single-month spend peak	₹2.19 trillion	March 2026
E-commerce share of March spend	₹1.41 trillion (64%)	March 2026
Market size	\$20.1 billion	2025
Projected market size	\$38.3 billion	2034 (CAGR 7.41%)

Appendix B: Sources

#	Claim	Source
1	India crossed 100M credit cards (March 2024); ~113M by Q3 2025	Business Standard: RBI Data
2	Average fintech onboarding drop-off is 63%	Eleken: Fintech Onboarding UX
3	68% of consumers abandoned a financial onboarding process	Tangle: Why Fintech Onboarding Flows Fail
4	70% of financial institutions lost clients due to inefficient onboarding (Fenergo, 2025)	Lorikeet: AI Onboarding in Fintech
5	Axis Bank Freecharge Credit Card details	Axis Bank Official
6	Axis Bank Freecharge Plus Credit Card details	Axis Bank Official
7	India credit card market 20.1B(2025), 38.3B by 2034, 7.41% CAGR	IMARC Group Market Report

8	UPI-linked credit card transactions = ~40% of all CC transactions by volume	GoKiwi: RBI & NPCI Data
9	118.63M cards, ₹2.19T March 2026 spend, ₹23.62T FY26 total, +12% YoY, e-commerce ₹1.41T	RBI Data: March 2026 (synthesised; verify against official RBI publications before final submission)